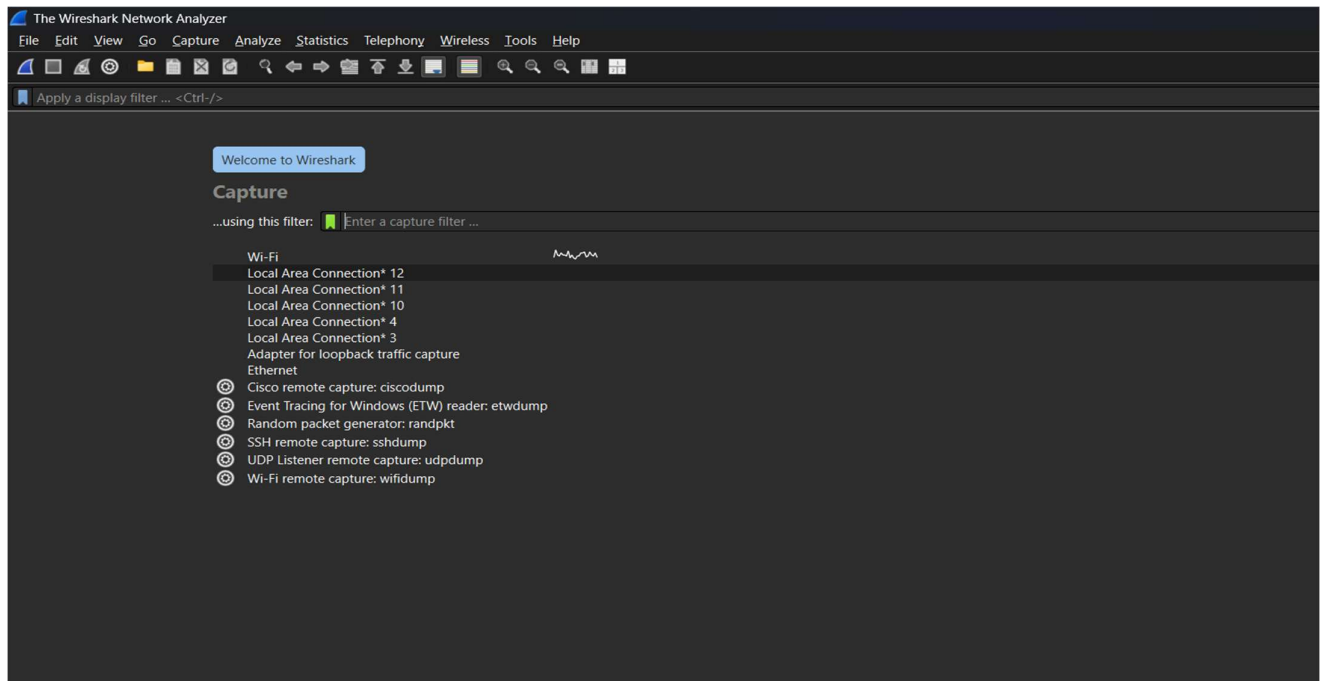


ASSIGNMENT 1

1. To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.

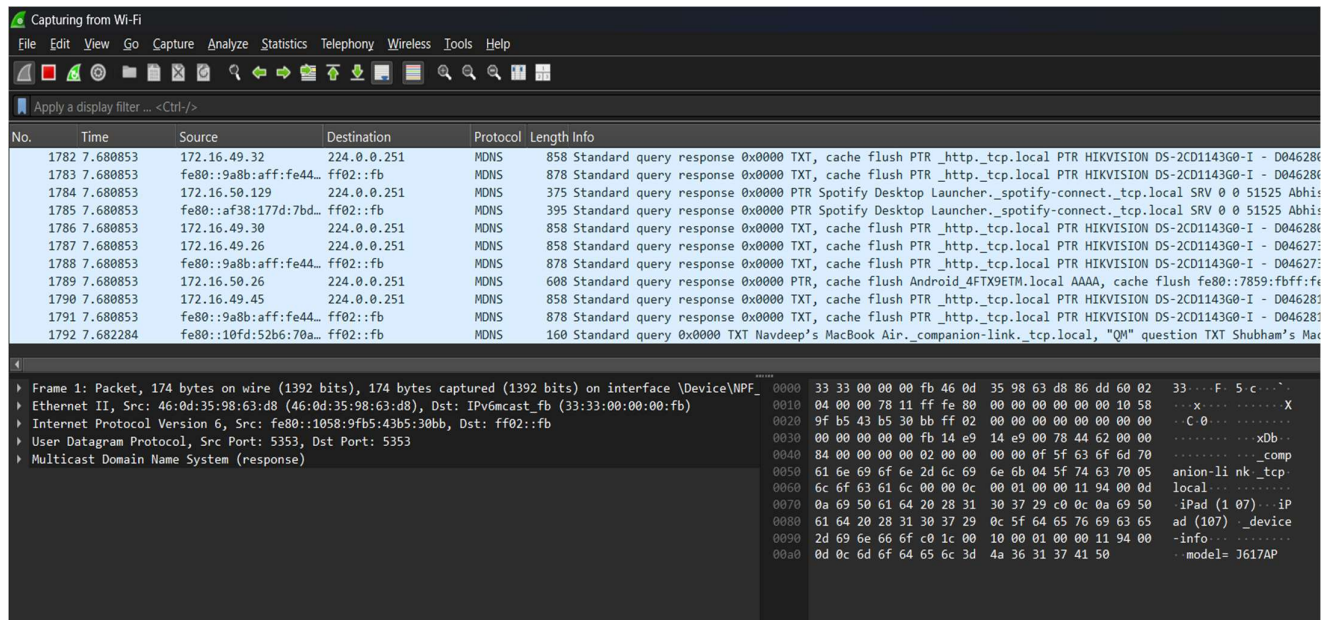
Step 1: Launch Wireshark

- Open **Wireshark**
- You will see a list of **network interfaces**
 - Wi-Fi
 - Ethernet



Step 2: Start Packet Capture

- Double-click on **Wi-Fi** (or active interface)
- Packet capture will start immediately



Step 3: Generate Network Traffic

Perform any of the following:

- Open a web browser
- Visit any website (e.g., google.com)
- Open Command Prompt and run:
- ping google.com

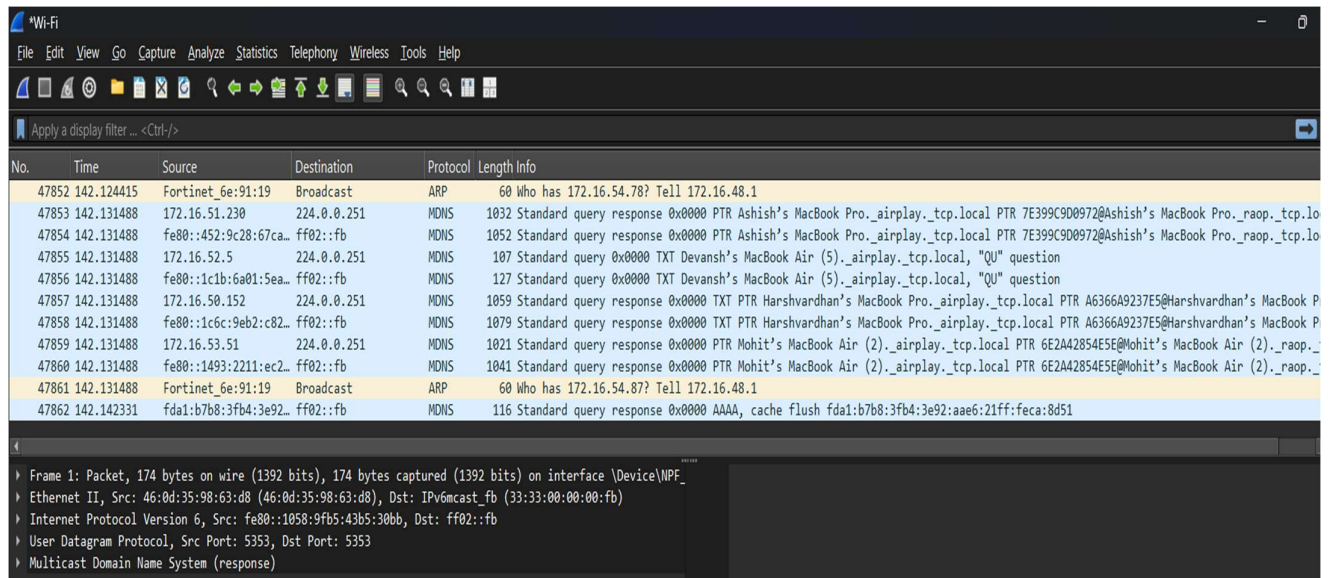
```
C:\Users\dell9>ping google.com

Pinging google.com [142.250.182.238] with 32 bytes of data:
Reply from 142.250.182.238: bytes=32 time=53ms TTL=118
Reply from 142.250.182.238: bytes=32 time=14ms TTL=118
Reply from 142.250.182.238: bytes=32 time=18ms TTL=118
Reply from 142.250.182.238: bytes=32 time=32ms TTL=118

Ping statistics for 142.250.182.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 53ms, Average = 29ms
```

Step 4: Stop Capture

- Click the red Stop button
- Captured packets will remain on the screen



Step 5: Apply Basic Filters

Use the **Display Filter** bar:

Filter	Purpose
ip	Show IP packets
tcp	Show TCP packets
udp	Show UDP packets
dns	Show DNS packets
icmp	Show ping packets

tcp					
No.	Time	Source	Destination	Protocol	Length Info
6892	13.725480	172.16.41.135	13.204.106.34	TCP	55 60059 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
6896	13.728040	13.204.106.34	172.16.41.135	TCP	66 443 → 60059 [ACK] Seq=1 Ack=2 Win=103 Len=0 SLE=1 SRE=2
7705	14.957805	172.16.41.135	74.125.200.188	TCP	55 58464 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1
7746	15.038603	74.125.200.188	172.16.41.135	TCP	66 5228 → 58464 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
8973	17.342393	172.16.41.135	52.111.224.7	TCP	66 56569 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8976	17.345076	52.111.224.7	172.16.41.135	TCP	66 443 → 56569 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256
8977	17.345148	172.16.41.135	52.111.224.7	TCP	54 56569 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
8979	17.346649	172.16.41.135	52.111.224.7	TLSv1.2	250 Client Hello (SNI=messaging.engagement.office.com)
8983	17.349960	52.111.224.7	172.16.41.135	TCP	60 443 → 56569 [ACK] Seq=1 Ack=197 Win=15872 Len=0
9196	17.701531	52.111.224.7	172.16.41.135	TLSv1.2	1514 Server Hello
9197	17.701531	52.111.224.7	172.16.41.135	TCP	1490 443 → 56569 [PSH, ACK] Seq=1461 Ack=197 Win=15872 Len=1436 [TCP PDU reassembled in 92]
9198	17.701630	172.16.41.135	52.111.224.7	TCP	54 56569 → 443 [ACK] Seq=197 Ack=2897 Win=65280 Len=0
9200	17.702894	52.111.224.7	172.16.41.135	TCP	1514 443 → 56569 [ACK] Seq=2897 Ack=197 Win=15872 Len=1460 [TCP PDU reassembled in 9203]
9201	17.702894	52.111.224.7	172.16.41.135	TCP	1490 443 → 56569 [PSH, ACK] Seq=4357 Ack=197 Win=15872 Len=1436 [TCP PDU reassembled in 92]
9203	17.702894	52.111.224.7	172.16.41.135	TLSv1.2	1502 Certificate
9204	17.702894	52.111.224.7	172.16.41.135	TLSv1.2	1514 Certificate Status
9205	17.702894	52.111.224.7	172.16.41.135	TLSv1.2	362 Server Key Exchange, Server Hello Done
9206	17.702961	172.16.41.135	52.111.224.7	TCP	54 56569 → 443 [ACK] Seq=197 Ack=5793 Win=65280 Len=0
9207	17.703110	172.16.41.135	52.111.224.7	TCP	54 56569 → 443 [ACK] Seq=197 Ack=9009 Win=65280 Len=0
9208	17.706135	172.16.41.135	52.111.224.7	TLSv1.2	180 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
▶ Frame 6892: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{0000 04 d5 90 6e 91 0f 98 43} fa 0a fe 60 08 00 45 00 ▶ Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f) ▶ Internet Protocol Version 4, Src: 172.16.41.135, Dst: 13.204.106.34 ▶ Transmission Control Protocol, Src Port: 60059, Dst Port: 443, Seq: 1, Ack: 1, Len: 1					

udp					
No.	Time	Source	Destination	Protocol	Length Info
6913	13.792808	fe80::cbc:18e:982f::	ff02::fb	MDNS	480 Standard query response 0x0000 PTR Gautam's MacBook Pro_companion-link_tcp.local TXT TXT, cache flush AAAA, cache flush fe80::
6914	13.792952	172.16.37.209	224.0.0.251	MDNS	456 Standard query response 0x0000 PTR Gautam's MacBook Pro_companion-link_tcp.local TXT TXT, cache flush AAAA, cache flush fe80::
6917	13.803743	172.16.41.135	173.194.154.38	UDP	75 51085 → 443 Len=33
6918	13.808004	172.16.40.74	224.0.0.251	MDNS	476 Standard query response 0x0000 PTR Bose Tech 11.1_companion-link_tcp.local TXT TXT, cache flush AAAA, cache flush fe80::1489::
6919	13.808486	fe80::1489:2cf:251::	ff02::fb	MDNS	500 Standard query response 0x0000 PTR Bose Tech 11.1_companion-link_tcp.local TXT TXT, cache flush AAAA, cache flush fe80::1489::
6921	13.815118	172.16.38.188	224.0.0.251	MDNS	100 Standard query 0x0000 PTR_rdlink_tcp.local, "QM" question PTR_companion-link_tcp.local, "QM" question
6922	13.815246	fe80::148f:fbb2:f3f::	ff02::fb	MDNS	124 Standard query 0x0000 PTR_rdlink_tcp.local, "QM" question PTR_companion-link_tcp.local, "QM" question
6923	13.820218	fe80::180c:b840:79d::	ff02::fb	MDNS	781 Standard query 0x0000 PTR_rdlink_tcp.local, "QM" question PTR_companion-link_tcp.local, "QM" question PTR Hardik's MacBook...
6924	13.820368	172.16.39.97	224.0.0.251	MDNS	757 Standard query 0x0000 PTR_rdlink_tcp.local, "QM" question PTR_companion-link_tcp.local, "QM" question PTR Hardik's MacBook...
6925	13.834393	fe80::855:e380:a211::	ff02::fb	MDNS	268 Standard query response 0x0000 TXT, cache flush NSEC, cache flush Air_companion-link_tcp.local
6926	13.834598	172.16.39.147	224.0.0.251	MDNS	244 Standard query response 0x0000 TXT, cache flush NSEC, cache flush Air_companion-link_tcp.local
6927	13.862722	fe80::2628:fdf:fef::	ff02::fb	MDNS	424 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31130507_https_tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU -
6928	13.863016	172.16.33.73	224.0.0.251	MDNS	400 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31130507_https_tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU -
6929	13.864682	fe80::2628:fdf:fef::	ff02::fb	MDNS	424 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459854_https_tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU -
6930	13.865402	172.16.33.94	224.0.0.251	MDNS	400 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459854_https_tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU -
6931	13.866607	173.194.154.38	172.16.41.135	UDP	1296 443 → 51085 Len=1250
6933	13.872134	172.16.41.135	173.194.154.38	UDP	75 51085 → 443 Len=33
6934	13.874120	fe80::2628:fdf:fef::	ff02::fb	MDNS	453 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459853_https_tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU -
6935	13.874120	fe80::7305:21e8:a91::	ff02::fb	MDNS	112 Standard query 0x0000 ANY Aditya(59) _dovsc_tcp.local, "QU" question
6936	13.874359	172.16.33.95	224.0.0.251	MDNS	429 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459853_https_tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU -
▶ Frame 6891: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF_{0000 04 d5 90 6e 91 0f 98 43} fa 0a fe 60 08 00 45 00 ▶ Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f) ▶ Internet Protocol Version 4, Src: 172.16.41.135, Dst: 173.194.154.38 ▶ User Datagram Protocol, Src Port: 51085, Dst Port: 443 ▶ Data (33 bytes)					

dns					
No.	Time	Source	Destination	Protocol	Length Info
3100	6.319183	172.16.41.135	172.31.1.6	DNS	70 Standard query 0xe0dd A google.com
3101	6.291474	172.16.41.135	172.31.1.6	DNS	70 Standard query 0xd228 AAAA google.com
3103	6.293499	172.31.1.6	172.16.41.135	DNS	86 Standard query response 0xe0dd A google.com A 172.217.24.238
3105	6.319569	172.31.1.6	172.16.41.135	DNS	98 Standard query response 0xd228 AAAA google.com AAAA 2404:6800:4002:82a::200e
8967	17.321777	172.16.41.135	172.31.1.6	DNS	91 Standard query 0x1d09 A messaging.engagement.office.com
8968	17.321967	172.16.41.135	172.31.1.6	DNS	91 Standard query 0x8464 AAAA messaging.engagement.office.com
8969	17.323584	172.31.1.6	172.16.41.135	DNS	182 Standard query response 0x1d09 A messaging.engagement.office.com CNAME prod-campaignagggregator.ome
8972	17.337200	172.31.1.6	172.16.41.135	DNS	194 Standard query response 0x8464 AAAA messaging.engagement.office.com CNAME prod-campaignagggregator.ome
9440	18.080831	172.16.41.135	172.31.1.6	DNS	86 Standard query 0xe85b A recent.svc.cloud.microsoft
9441	18.080967	172.16.41.135	172.31.1.6	DNS	86 Standard query 0xe3c2 AAAA recent.svc.cloud.microsoft
9444	18.084769	172.31.1.6	172.16.41.135	DNS	368 Standard query response 0x853b A recent.svc.cloud.microsoft CNAME prod.ocws1.live.com.akadns.net C
9445	18.110026	172.16.41.135	8.8.8.8	DNS	86 Standard query 0xe3c2 AAAA recent.svc.cloud.microsoft
9449	18.139246	172.31.1.6	172.16.41.135	DNS	464 Standard query response 0xe3c2 AAAA recent.svc.cloud.microsoft CNAME prod.ocws1.live.com.akadns.net
9492	18.257844	172.16.41.135	172.31.1.6	DNS	80 Standard query 0x32c2 A substrate.office.com
9493	18.258247	172.16.41.135	172.31.1.6	DNS	80 Standard query 0x4e9f AAAA substrate.office.com
9495	18.270105	172.31.1.6	172.16.41.135	DNS	266 Standard query response 0x32c2 A substrate.office.com CNAME outlook.office365.com CNAME ooc-g2.tn-
9500	18.283716	172.31.1.6	172.16.41.135	DNS	302 Standard query response 0x4e9f AAAA substrate.office.com CNAME outlook.office365.com CNAME ooc-g2.tn-
9501	18.283834	172.16.41.135	8.8.8.8	DNS	80 Standard query 0x4e9f AAAA substrate.office.com
9515	18.300042	8.8.8.8	172.16.41.135	DNS	302 Standard query response 0x4e9f AAAA substrate.office.com CNAME outlook.office365.com CNAME ooc-g2.tn-
10048	19.497391	172.16.41.135	172.31.1.6	DNS	86 Standard query 0x5279 A editor.svc.cloud.microsoft
▶ Frame 3105: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface \Device\NPF_{0000 04 d5 90 6e 91 0f 98 43} fa 0a fe 60 08 00 45 00 ▶ Ethernet II, Src: Fortinet_6e:91:0f (04:d5:90:6e:91:0f), Dst: Intel_0a:fe:60 (98:43:fa:0a:fe:60) ▶ Internet Protocol Version 4, Src: 172.31.1.6, Dst: 172.16.41.135 ▶ User Datagram Protocol, Src Port: 53, Dst Port: 50387 ▶ Domain Name System (response)					

No.	icmp	Source	Destination	Protocol	Length	Info
→ 3111	6.3805499	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 3127)
← 3127	6.380559	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=118 (request in 3111)
3513	7.342610	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 3518)
3518	7.364507	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=118 (request in 3513)
4290	8.357826	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 4323)
4323	8.394766	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=118 (request in 4290)
4906	9.374616	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 4911)
4911	9.402421	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=118 (request in 4906)

Step 6: Analyze a Packet

- Click on any packet
- Observe:
 - Ethernet header
 - IP header
 - TCP/UDP header
 - Payload data

Wireshark - Packet 4906 - Wi-Fi

Frame 4906: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0

Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)

Destination: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)

Source: Intel_0a:fe:60 (98:43:fa:0a:fe:60)

Type: IPv4 (0x0800)

[Stream index: 4]

Internet Protocol Version 4, Src: 172.16.41.135, Dst: 172.217.24.238

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 60

Identification: 0x2c90 (11408)

000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 128

Protocol: ICMP (1)

Header Checksum: 0x0000 [validation disabled]

[Header checksum status: Unverified]

Source Address: 172.16.41.135

Destination Address: 172.217.24.238

[Stream index: 177]

Internet Control Message Protocol

```

0000  04 d5 90 6e 91 0f 98 43 fa 0a fe 60 08 00 45 00  ...n..C...E
0010  00 3c 2c 90 00 00 80 01 00 00 ac 10 29 87 ac d9  <,.....)
0020  18 ee 08 00 4d 57 00 01 00 04 61 62 63 64 65 66  ...MW...abcdef
0030  67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76  ghijklmn opqrstuv
0040  77 61 62 63 64 65 66 67 68 69                    wabcdefg hi
  
```

Observation

Interface used:

- Protocols observed:
- Source and destination IP addresses
- Packet length

Result

Live network packets were successfully captured using Wireshark, and basic network protocols were identified and analyzed.

ASSIGNMENT 2

1. To capture and analyze ICMP packets (Ping request and reply)

```
C:\Users\dell9>ping google.com

Pinging google.com [172.217.24.238] with 32 bytes of data:
Reply from 172.217.24.238: bytes=32 time=46ms TTL=118
Reply from 172.217.24.238: bytes=32 time=22ms TTL=118
Reply from 172.217.24.238: bytes=32 time=37ms TTL=118
Reply from 172.217.24.238: bytes=32 time=27ms TTL=118

Ping statistics for 172.217.24.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 46ms, Average = 33ms
```

No.	icmp icmpv6	Source	Destination	Protocol	Length	Info
3111 6.394199		172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 3127)
3127 6.380559		172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=118 (request in 3111)
3513 7.342610		172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 3518)
3518 7.364507		172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=118 (request in 3513)
4290 8.357826		172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 4323)
4323 8.394766		172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=118 (request in 4290)
→ 4906 9.374616		172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 4911)
← 4911 9.402421		172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=118 (request in 4906)

2. To analyze TCP packets and observe the TCP three-way handshake

No.	Time	Source	Destination	Protocol	Length	Info
6892	13.725480	172.16.41.135	13.204.106.34	TCP	55	60059 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
6893	13.728040	172.16.41.135	13.204.106.34	TCP	66	443 → 60059 [ACK] Seq=1 Ack=2 Win=193 Len=0 SLE=1 SRE=2
7705	14.957805	172.16.41.135	74.125.209.188	TCP	55	58464 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=0
7746	15.038603	74.125.209.188	172.16.41.135	TCP	66	5228 → 58464 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
8973	17.342393	172.16.41.135	52.111.224.7	TCP	66	56569 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8976	17.345076	52.111.224.7	172.16.41.135	TCP	66	443 → 56569 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256
8977	17.345148	172.16.41.135	52.111.224.7	TCP	54	56569 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
8979	17.346469	172.16.41.135	52.111.224.7	TLSv1.2	250	Client Hello (SN=messaging.engagement.office.com)
8983	17.349606	52.111.224.7	172.16.41.135	TCP	60	443 → 56569 [ACK] Seq=1 Ack=197 Win=15872 Len=0
9197	17.701537	172.16.41.135	172.16.41.135	TLSv1.2	1514	Server Hello
9197	17.701531	52.111.224.7	172.16.41.135	TCP	1490	443 → 56569 [PSH, ACK] Seq=1461 Ack=197 Win=15872 Len=1436 [TCP PDU reassembled in 9203]
9198	17.701630	172.16.41.135	52.111.224.7	TCP	54	56569 → 443 [ACK] Seq=197 Ack=2897 Win=65280 Len=0
9200	17.702894	52.111.224.7	172.16.41.135	TCP	1514	443 → 56569 [ACK] Seq=2897 Ack=197 Win=15872 Len=1460 [TCP PDU reassembled in 9203]
9201	17.702894	52.111.224.7	172.16.41.135	TCP	1490	443 → 56569 [PSH, ACK] Seq=4357 Ack=197 Win=15872 Len=1436 [TCP PDU reassembled in 9203]
9203	17.702894	52.111.224.7	172.16.41.135	TLSv1.2	1502	Certificate
9204	17.702894	52.111.224.7	172.16.41.135	TLSv1.2	1514	Certificate Status
9205	17.702897	52.111.224.7	172.16.41.135	TLSv1.2	362	Server Key Exchange, Server Hello Done
9206	17.702961	172.16.41.135	52.111.224.7	TCP	54	56569 → 443 [ACK] Seq=197 Ack=5793 Win=65280 Len=0
9207	17.703110	172.16.41.135	52.111.224.7	TCP	54	56569 → 443 [ACK] Seq=197 Ack=9009 Win=65280 Len=0
9208	17.706135	172.16.41.135	52.111.224.7	TLSv1.2	180	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
9209	17.708115	52.111.224.7	172.16.41.135	TCP	60	443 → 56569 [ACK] Seq=9009 Ack=323 Win=15872 Len=0
9363	17.883072	52.111.224.7	172.16.41.135	TLSv1.2	328	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
9372	17.891479	172.16.41.135	52.111.224.7	TLSv1.2	2313	Application Data
9374	17.901531	52.111.224.7	172.16.41.135	TCP	60	443 → 56569 [ACK] Seq=9283 Ack=1783 Win=18688 Len=0
9377	17.901537	52.111.224.7	172.16.41.135	TCP	60	443 → 56569 [ACK] Seq=9283 Ack=2582 Win=18688 Len=0
9451	18.140473	172.16.41.135	52.110.14.146	TCP	66	56570 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
9452	18.140826	172.16.41.135	52.110.14.146	TCP	66	56571 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
9454	18.146720	52.110.14.146	172.16.41.135	TCP	66	443 → 56570 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256
9455	18.146720	52.110.14.146	172.16.41.135	TCP	66	443 → 56571 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256

```

Wireshark - Packet 8976 - Wi-Fi

> Frame 8976: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0
> Ethernet II, Src: Fortinet 6e:91:0f (04:d5:90:6e:91:0f), Dst: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
> Internet Protocol Version 4, Src: 52.111.224.7, Dst: 172.16.41.135
> Transmission Control Protocol, Src Port: 443, Dst Port: 56569, Seq: 0, Ack: 1, Len: 0
  Source Port: 443
  Destination Port: 56569
  [Stream index: 2]
  [Stream Packet Number: 2]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 1333957722
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 1272899674
  1000 .... = Header Length: 32 bytes (8)
  Flags: 0x012 (SYN, ACK)
  Window: 14600
  [Calculated window size: 14600]
  Checksum: 0x551f [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operation (NOP), Window scale
  [Timestamps]
  [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]

0000  98 43 fa 0a fe 60 04 d5 90 6e 91 0f 08 00 45 00  .C...n...E-
0010  00 34 f9 1f 40 00 06 57 96 34 6f e0 07 ac 10  -4.@W4o...
0020  29 87 01 bb cf f9 4f 82 94 5a 4b de e8 5a 80 12  )...O-ZK-Z-
0030  39 08 55 1f 00 00 02 04 05 b4 01 01 04 02 01 03  9-U.....
0040  03 08                                     +..

```

3. To capture and analyze UDP packets

The image displays a Wireshark packet capture analysis of a UDP packet. The top pane shows a list of network packets, with packet 8989 selected. The middle pane shows the packet details for Ethernet II, 802.1Q Virtual LAN, Internet Protocol Version 4, and User Datagram Protocol. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
8968	17.321967	172.16.41.135	172.31.1.6	DNS	91	Standard query 0x8464 AAAA messaging.engagement.office.com
8969	17.323584	172.31.1.6	172.16.41.135	DNS	182	Standard query response 0xd09 A messaging.engagement.office.com CNAME prod-campaignaggregator.omexexternalfb.office.net.akad.
8970	17.331977	173.194.154.38	172.16.41.135	UDP	1296	443 → 51885 Len=1250
8972	17.337200	172.31.1.6	172.16.41.135	DNS	194	Standard query response 0x8464 AAAA messaging.engagement.office.com CNAME prod-campaignaggregator.omexexternalfb.office.net.a.
8974	17.344875	142.250.195.46	172.16.41.135	UDP	70	443 → 59855 Len=24
8975	17.345072	172.16.41.135	142.250.195.46	UDP	73	59855 → 443 Len=31
8978	17.345165	172.16.41.135	173.194.154.38	UDP	75	51885 → 443 Len=33
8980	17.347519	fe80::aaf:89ff:fe55::ff02::fb	224.0.0.251	MDNS	424	Standard query 0x0000 ANY HIKVISION DS-2CD3651G0-IZ - F60580171._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3651G0-IZ
8982	17.347888	172.16.33.33	224.0.0.251	MDNS	400	Standard query 0x0000 ANY HIKVISION DS-2CD3651G0-IZ - F60580171._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3651G0-IZ
8986	17.382648	172.16.33.82	224.0.0.251	MDNS	429	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459834._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU
8987	17.383073	fe80::2628:fdff:fef::ff02::fb	224.0.0.251	MDNS	453	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459834._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU
8988	17.388697	142.250.195.46	172.16.41.135	UDP	114	443 → 59855 Len=68
8989	17.397498	142.250.195.46	172.16.41.135	UDP	67	443 → 59855 Len=21
8990	17.397831	172.16.41.135	142.250.195.46	UDP	77	59855 → 443 Len=35
8991	17.412490	fe80::2628:fdff:fec::ff02::fb	224.0.0.251	MDNS	424	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31130577._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU
8992	17.412668	172.16.33.79	224.0.0.251	MDNS	400	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31130577._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU
8993	17.413591	173.194.154.38	172.16.41.135	UDP	1296	443 → 51885 Len=1250
8994	17.418481	172.16.36.131	224.0.0.251	MDNS	349	Standard query response 0x0000 PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local
8995	17.419113	fe80::1805:a078:e19::ff02::fb	224.0.0.251	MDNS	373	Standard query response 0x0000 PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local
8996	17.419690	172.16.36.119	224.0.0.251	MDNS	1047	Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question PTR Akshin's iPad (2.
8997	17.421600	fe80::1c52:6de9:a09::ff02::fb	224.0.0.251	MDNS	1071	Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question PTR Akshin's iPad (2.
8998	17.423066	172.16.33.86	224.0.0.251	MDNS	429	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459843._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU
8999	17.423962	fe80::2628:fdff:fef::ff02::fb	224.0.0.251	MDNS	453	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459843._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU
9000	17.426562	fe80::2628:fdff:fef::ff02::fb	224.0.0.251	MDNS	453	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459850._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU
9001	17.426789	172.16.33.80	224.0.0.251	MDNS	429	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459850._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU
9002	17.428458	172.16.41.135	173.194.154.38	UDP	75	51885 → 443 Len=33

Frame 8989: Packet, 67 bytes on wire (536 bits), 67 bytes captured (536 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0

Ethernet II, Src: Fortinet_6e:91:0f (04:d5:90:6e:91:0f), Dst: Intel_0a:fe:60 (98:43:fa:0a:fe:60)

Destination: Intel_0a:fe:60 (98:43:fa:0a:fe:60)

Source: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)

Type: 802.1Q Virtual LAN (0x8100)

[Stream index: 4]

802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0

000. = Priority: Best Effort (default) (0)

...0 = DEI: Ineligible

.... 0000 0000 0000 = ID: 0

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 142.250.195.46, Dst: 172.16.41.135

User Datagram Protocol, Src Port: 443, Dst Port: 59855

Source Port: 443

Destination Port: 59855

Length: 29

Checksum: 0xe883 [unverified]

[Checksum Status: Unverified]

[Stream index: 368]

[Stream Packet Number: 15]

[Timestamps]

UDP payload (21 bytes)

Data (21 bytes)

0000 98 43 fa 0a fe 60 04 d5 90 6e 91 0f 81 00 00 00 C n

0010 08 00 45 80 00 31 00 00 40 00 3c 11 16 7c 8e fa . E . . . @ < . | . .

0020 c3 2e ac 10 29 87 01 bb e9 cf 00 1d e8 83 50 1a) P

0030 82 10 87 bc 1e e5 46 88 63 93 23 f6 2d 21 0a 13 F c # - | . .

0040 e9 d1 9c

No: 8989 - Time: 17.397498 - Source: 142.250.195.46 - Destination: 172.16.41.135 - Protocol: UDP - Length: 67 - Info: 443 → 59855 Len=21

4. To capture and analyze DNS query and response packets

dns						
No.	Time	Source	Destination	Protocol	Length	Info
71950	92.027577	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x0003 A fa000000191.resources.office.net
16846	31.105719	172.16.41.135	172.31.1.6	DNS	85	Standard query 0x03bf A fd.api.iris.microsoft.com
75735	92.889790	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x04e6 A fa000000193.resources.office.net
75773	92.908395	172.16.41.135	8.8.8.8	DNS	92	Standard query 0x04e6 A fa000000193.resources.office.net
17123	31.660998	172.16.41.135	172.31.1.6	DNS	78	Standard query 0x0ec8 AAAA uci.cdn.office.net
64804	90.185509	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x0eed A fa000000145.resources.office.net
66245	90.652437	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x15eb A fa000000175.resources.office.net
8967	17.321777	172.16.41.135	172.31.1.6	DNS	91	Standard query 0x1d09 A messaging.engagement.office.com
45655	85.668091	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x278d A fa000000128.resources.office.net
54196	87.491990	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x293c A fa000000136.resources.office.net
44264	85.236578	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x2c65 A fa000000124.resources.office.net
30496	57.952915	172.16.41.135	172.31.1.6	DNS	87	Standard query 0x2dd3 AAAA roaming.svc.cloud.microsoft
30521	57.983417	172.16.41.135	8.8.8.8	DNS	87	Standard query 0x2dd3 AAAA roaming.svc.cloud.microsoft
14265	27.005871	172.16.41.135	172.31.1.6	DNS	90	Standard query 0x2e5c AAAA self.events.data.microsoft.com
14264	27.005784	172.16.41.135	172.31.1.6	DNS	90	Standard query 0x31bd A self.events.data.microsoft.com
9492	18.257844	172.16.41.135	172.31.1.6	DNS	80	Standard query 0x32c2 A substrate.office.com
42619	82.146241	172.16.41.135	172.31.1.6	DNS	80	Standard query 0x3535 AAAA cs.dds.microsoft.com
13718	26.441137	172.16.41.135	172.31.1.6	DNS	76	Standard query 0x391e A fs.microsoft.com
66257	90.654212	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x3c3a AAAA fa000000156.resources.office.net
66249	90.652830	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x3c74 AAAA fa000000175.resources.office.net
13719	26.441236	172.16.41.135	172.31.1.6	DNS	76	Standard query 0x3ee1 AAAA fs.microsoft.com

```
Wireshark - Packet 45655 - Wi-Fi
  Frame 45655: Packet, 92 bytes on wire (736 bits), 92 bytes captured (736 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0
  Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
    Destination: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
    Source: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
    Type: IPv4 (0x0800)
    [Stream index: 4]
  Internet Protocol Version 4, Src: 172.16.41.135, Dst: 172.31.1.6
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 78
    Identification: 0xffd5 (65493)
    000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.16.41.135
    Destination Address: 172.31.1.6
    [Stream index: 176]
  User Datagram Protocol, Src Port: 64655, Dst Port: 53
    Source Port: 64655
    Destination Port: 53
    Length: 58
    Checksum: 0x8308 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 539]
    [Stream Packet Number: 23]
    [Timestamps]
    UDP payload (50 bytes)
  Domain Name System (query)
```

5. To capture and analyze HTTP packets

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
23790	53.450858	172.16.41.135	34.223.124.45	HTTP	493	GET / HTTP/1.1

▼ Frame 23790: Packet, 493 bytes on wire (3944 bits), 493 bytes captured (3944 bits) on interface 'Device'

▼ Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)

 Destination: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)

 Source: Intel_0a:fe:60 (98:43:fa:0a:fe:60)

 Type: IPv4 (0x0800)

 [Stream index: 6]

▼ Internet Protocol Version 4, Src: 172.16.41.135, Dst: 34.223.124.45

 0100 = Version: 4

 0101 = Header Length: 20 bytes (5)

 ▼ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

 Total Length: 479

 Identification: 0x8662 (34402)

 ▼ 010 = Flags: 0x2, Don't fragment

 ...0 0000 0000 0000 = Fragment Offset: 0

 Time to live: 128

 Protocol: TCP (6)

 Header Checksum: 0x0000 [validation disabled]

 [Header checksum status: Unverified]

 Source Address: 172.16.41.135

 Destination Address: 34.223.124.45

 [Stream index: 358]

▼ Transmission Control Protocol, Src Port: 55322, Dst Port: 80, Seq: 1, Ack: 1, Len: 439

 Hypertext Transfer Protocol

```

0000  04 d5 90 6e 91 0f 98 43 fa 0a fe 60 08 00 45 00  ....C.....E
0010  01 df 86 62 40 00 80 06 00 00 ac 10 29 87 22 df  ..t.....)
0020  7c 2d d8 1a 00 50 d3 38 7e 3c b9 0e 8d c9 50 18  ...P8<...P
0030  00 ff 76 75 00 00 47 45 54 20 2f 20 48 54 54 50  ...vGE T / HTTP
0040  2f 31 2e 21 6d 0a 48 6f 73 7a 3a 20 6e 65 76 65  /1.1-Host: neve
0050  27 73 73 6c 2e 63 6f 6d 0d 0a 43 6f 6e 6e 65 63  rssl.com:Connec
0060  74 69 6f 6e 3a 20 6b 65 65 70 72 61 6c 69 69 65  tion: ke-apaliv
0070  0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63 75  Upgrad e-Instru
0080  72 65 52 65 61 75 65 73 74 73 3a 20 31 8d 0a 6f  -Request state: 1
0090  55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 4f 7a 69  User-Agent: Mozi
00a0  6c 6c 61 2f 35 20 30 20 28 57 69 6e 64 6f 77 73  lla/5.0 (Windows
00b0  20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 66 3a 34  NT 10.0; Win64;
00c0  20 78 36 34 29 20 41 70 70 6c 65 57 65 62 4b 69  x64) Ap pleWebKi
00d0  74 2f 35 33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c  t/537.36 (KHTML,
00e0  28 6c 69 6b 65 20 47 65 63 6b 6f 29 28 43 68 72  like Ge cko) Chr
00f0  6f 6d 65 2f 31 34 3a 2e 30 2e 30 2e 30 20 53 61  om/144, 0.0.0 Sa
0100  6e 61 72 69 2f 35 37 2e 33 36 0d 0a 41 63 63 6f  fari/537.36; Acc
0110  65 70 74 3a 20 65 74 78 74 2f 68 74 6d 6c 2c 61  ept: tex t/html,a
0120  70 78 6d 63 61 74 69 6f 6f 6e 2f 78 68 74 6d 6c  pplicati on/xhtml,a
0130  20 7b 6d 6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e  +xml,app lication
0140  2f 78 6d 6c 6b 37 3d 30 2e 39 2c 69 6d 61 67 65  /xml;q=0 .9,image
0150  2f 61 76 69 66 2c 69 6d 61 67 65 2f 77 65 62 70  /avif,im age/webp
0160  2c 69 6d 61 67 65 2f 61 70 6e 67 6c 2a 2f 2a 3b  ,image/a png; /*;
0170  71 3d 30 2e 39 2c 61 70 6c 69 63 61 74 69 6f 6e  q=0.8, ap plicatio
0180  6e 6f 30 68 61 67 65 74 78 68 69 63 61 67 6e 78  6e 6f 30 68 61 67

```


6. To analyze ARP request and reply packets

arp					
No.	Time	Source	Destination	Protocol	Length Info
5818	11.297199	Intel_0a:fe:60	Fortinet_6e:91:0f	ARP	42 172.16.41.135 is at 98:43:fa:0a:fe:60
5819	11.297227	Intel_0a:fe:60	Fortinet_6e:91:0f	ARP	42 172.16.41.135 is at 98:43:fa:0a:fe:60
35994	69.430284	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64 Who has 172.16.32.19? Tell 172.16.32.1
4425	8.559894	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.1? Tell 172.16.32.167
12753	24.701997	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.1? Tell 172.16.32.167
28969	54.727888	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.1? Tell 172.16.32.167
35975	69.275587	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.1? Tell 172.16.32.167
44039	84.753305	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.1? Tell 172.16.32.167
95972	114.769454	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.1? Tell 172.16.32.167
103583	128.846928	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.1? Tell 172.16.32.167
15400	29.354621	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.229? Tell 172.16.32.167
16488	30.357871	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.229? Tell 172.16.32.167
16959	31.358913	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.229? Tell 172.16.32.167
17619	32.361150	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.229? Tell 172.16.32.167
18257	33.362660	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.229? Tell 172.16.32.167
18732	34.365862	RuckusWirele_35:bd:...	Intel_0a:fe:60	ARP	46 Who has 172.16.32.229? Tell 172.16.32.167
44186	85.025542	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64 Who has 172.16.32.82? Tell 172.16.32.1
21987	40.777993	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64 Who has 172.16.36.102? Tell 172.16.32.1
22643	41.778668	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64 Who has 172.16.36.102? Tell 172.16.32.1
23066	42.778969	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64 Who has 172.16.36.102? Tell 172.16.32.1
41616	80.473221	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64 Who has 172.16.36.102? Tell 172.16.32.1

Wireshark - Packet 18257 - Wi-Fi	
Frame 18257: Packet, 46 bytes on wire (368 bits), 46 bytes captured (368 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0	
Ethernet II, Src: RuckusWirele_35:bd:00 (d8:38:fc:35:bd:00), Dst: Intel_0a:fe:60 (98:43:fa:0a:fe:60)	
Destination: Intel_0a:fe:60 (98:43:fa:0a:fe:60)	
Source: RuckusWirele_35:bd:00 (d8:38:fc:35:bd:00)	
Type: 802.1Q Virtual LAN (0x8100)	
[Stream index: 228]	
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0	
000. = Priority: Best Effort (default) (0)	
...0 = DEI: Ineligible	
.... 0000 0000 0000 = ID: 0	
Type: ARP (0x0806)	
Address Resolution Protocol (request)	
Hardware type: Ethernet (1)	
Protocol type: IPv4 (0x0800)	
Hardware size: 6	
Protocol size: 4	
Opcode: request (1)	
Sender MAC address: RuckusWirele_35:bd:00 (d8:38:fc:35:bd:00)	
Sender IP address: 172.16.32.167	
Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)	
Target IP address: 172.16.32.229	
0000	98 43 fa 0a fe 60 d8 38 fc 35 bd 00 81 00 00 00 C... 8 .5
0010	08 06 00 01 08 00 06 00 01 d8 38 fc 35 bd 00 8 .5
0020	ac 10 20 a7 00 00 00 00 00 00 ac 10 20 e5

7. To identify source and destination MAC and IP addresses in captured packets.

The image shows a Wireshark packet capture analysis. The top pane displays a list of captured packets, with packet 6892 selected. The middle pane shows the details of the selected packet, and the bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
6892	13.725480	172.16.41.135	13.204.106.34	TCP	55	60059 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
6896	13.728040	13.204.106.34	172.16.41.135	TCP	66	443 → 60059 [ACK] Seq=1 Ack=2 Win=103 Len=0 SLE=1 SRE=2
7705	14.957805	172.16.41.135	74.125.200.188	TCP	55	58464 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1
7746	15.038603	74.125.200.188	172.16.41.135	TCP	66	5228 → 58464 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2

Packet 6892 Details:

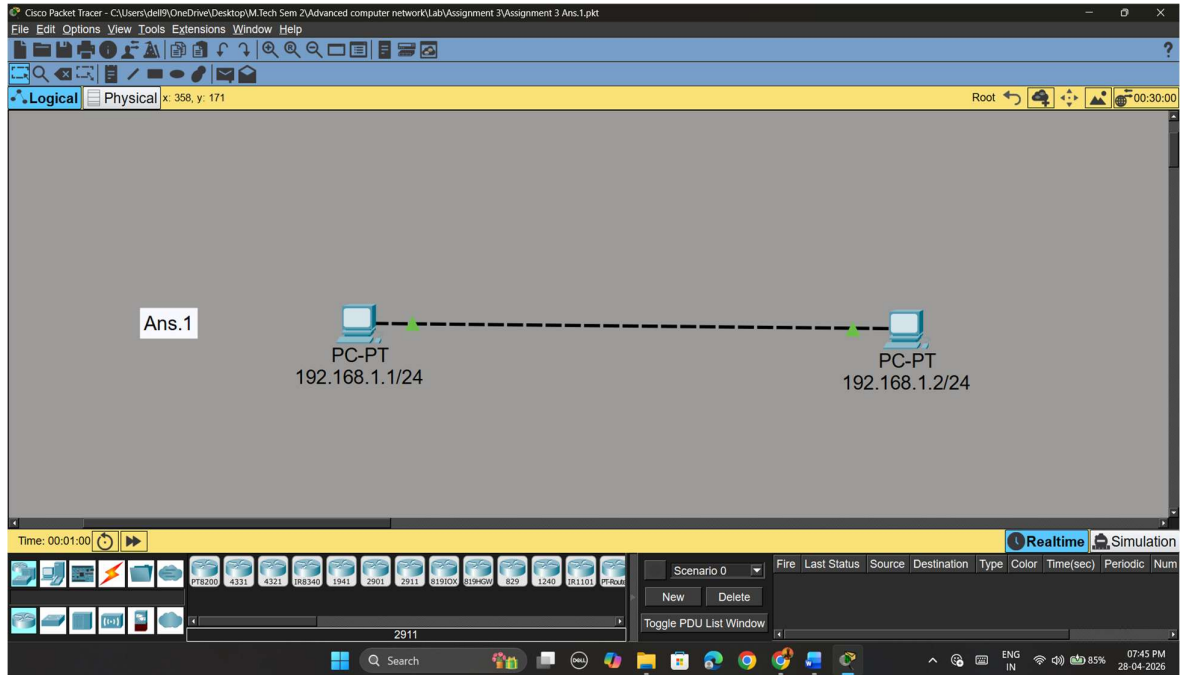
- Frame 6892: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{...}
- Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
 - Destination: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
 - Source: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
 - Type: IPv4 (0x0800)
 - [Stream index: 4]
- Internet Protocol Version 4, Src: 172.16.41.135, Dst: 13.204.106.34
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 41
 - Identification: 0xba38 (47672)
 - 010. = Flags: 0x2, Don't fragment
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 128
 - Protocol: TCP (6)
 - Header Checksum: 0x0000 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 172.16.41.135
 - Destination Address: 13.204.106.34
 - [Stream index: 213]
- Transmission Control Protocol, Src Port: 60059, Dst Port: 443, Seq: 1, Ack: 1, Len: 1

Raw Packet Data:

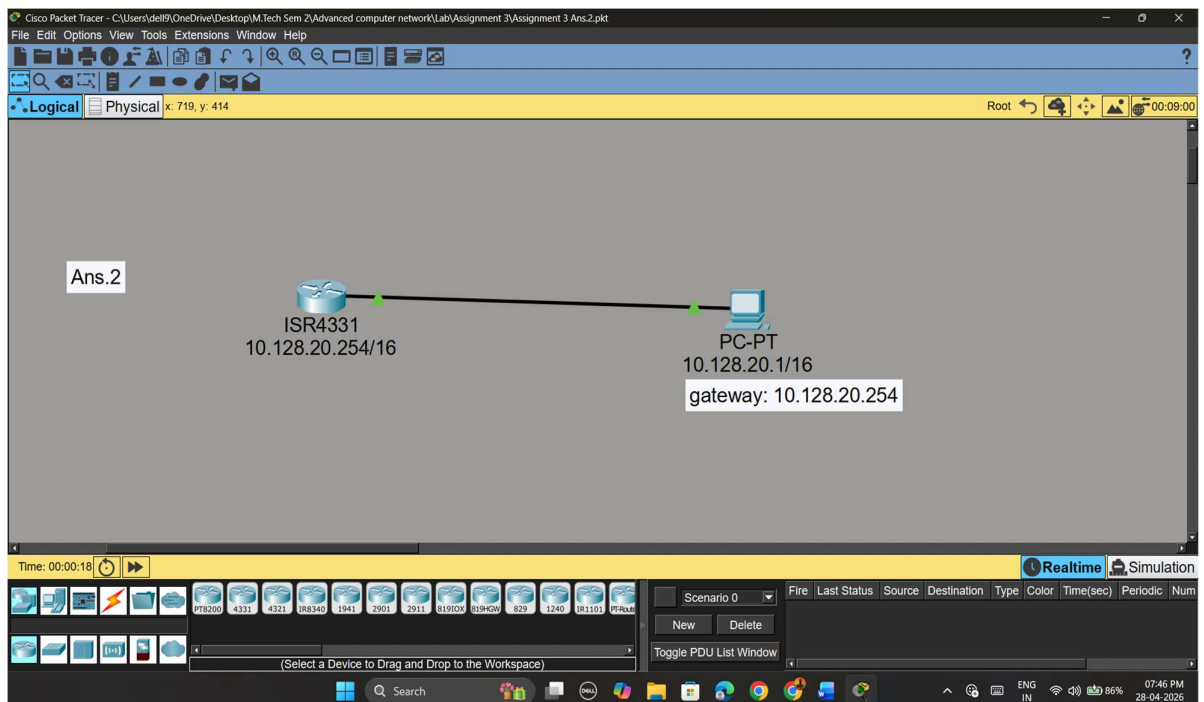
```
0000 04 d5 90 6e 91 0f 98 43 fa 0a fe 60 08 00 45 00 ... n...C
0010 00 29 ba 38 40 00 80 06 00 00 ac 10 29 87 0d cc ...) 8@ ...
0020 6a 22 ea 9b 01 bb bb 88 e8 46 c2 18 33 d4 50 10 j".....
0030 00 fc 4d a1 00 00 00
```

ASSIGNMENT 3

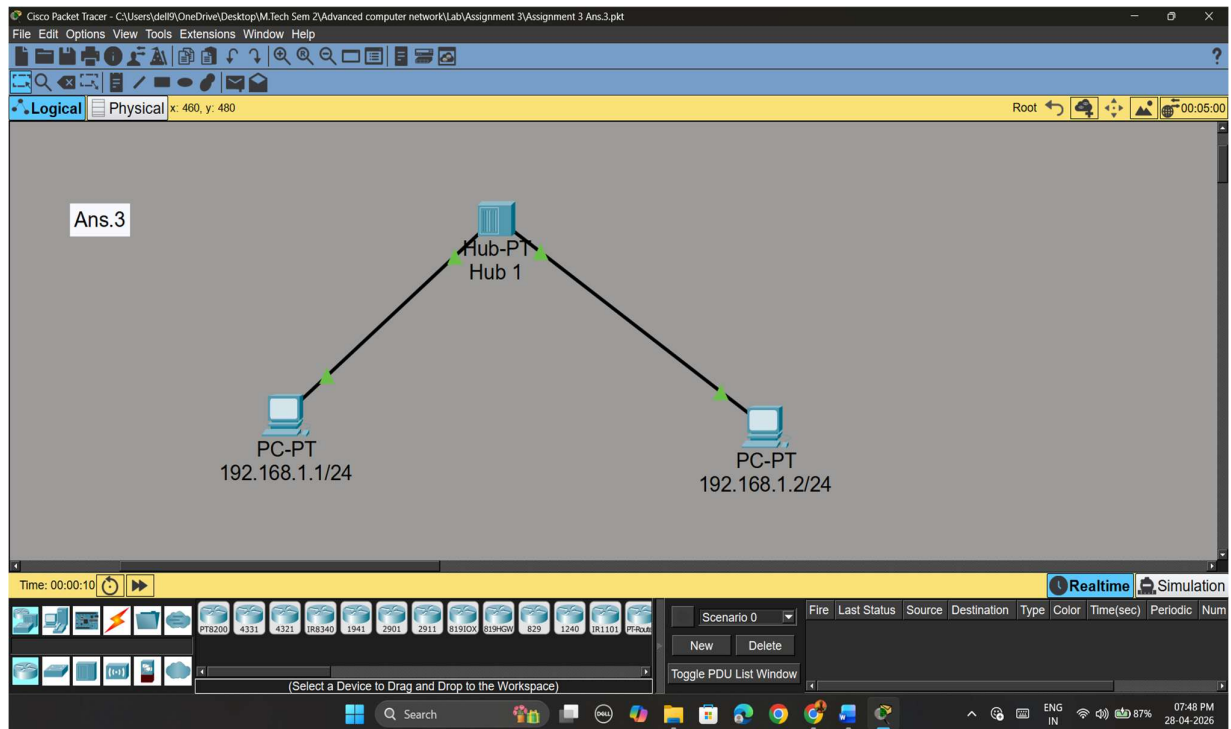
Ans.1



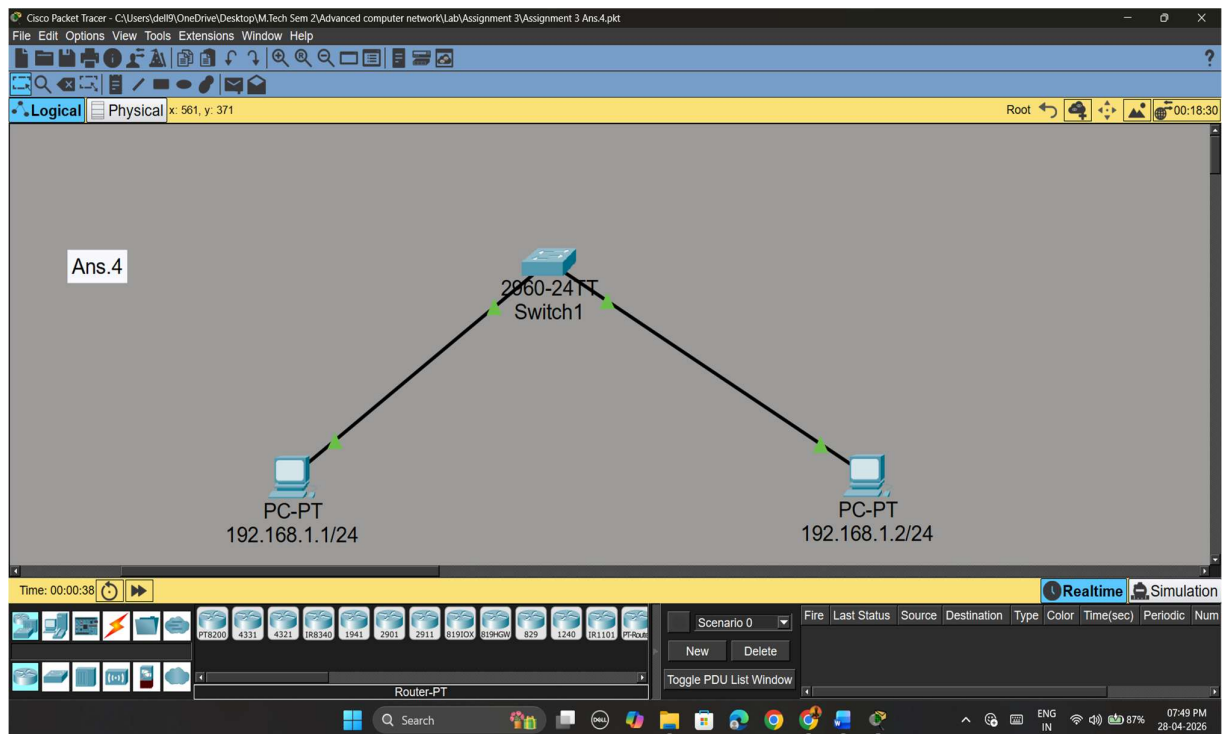
Ans.2



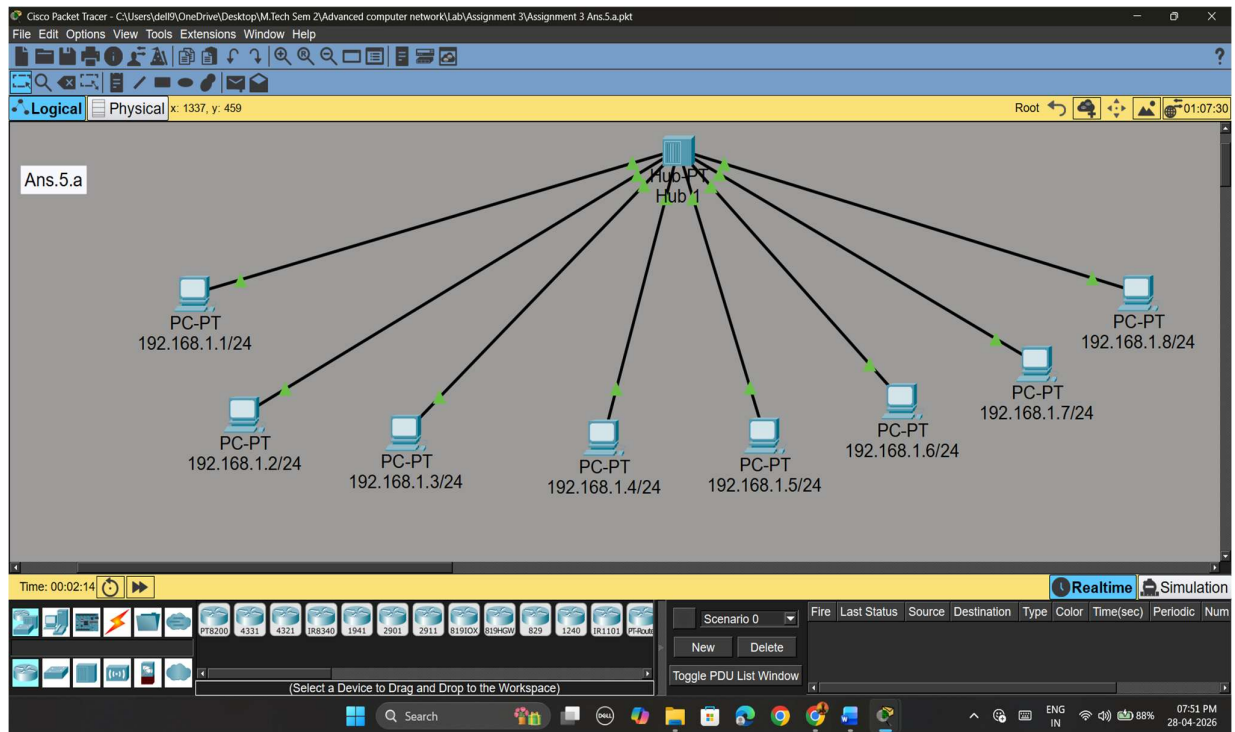
Ans.3



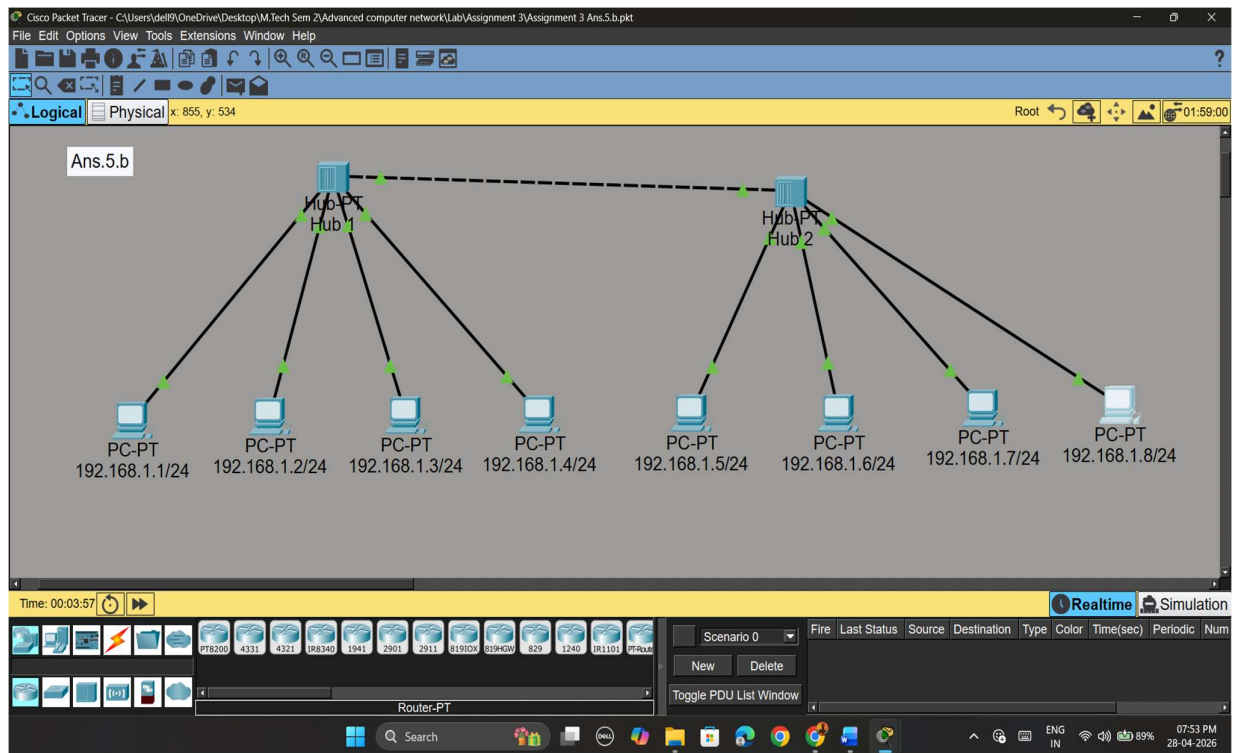
Ans.4



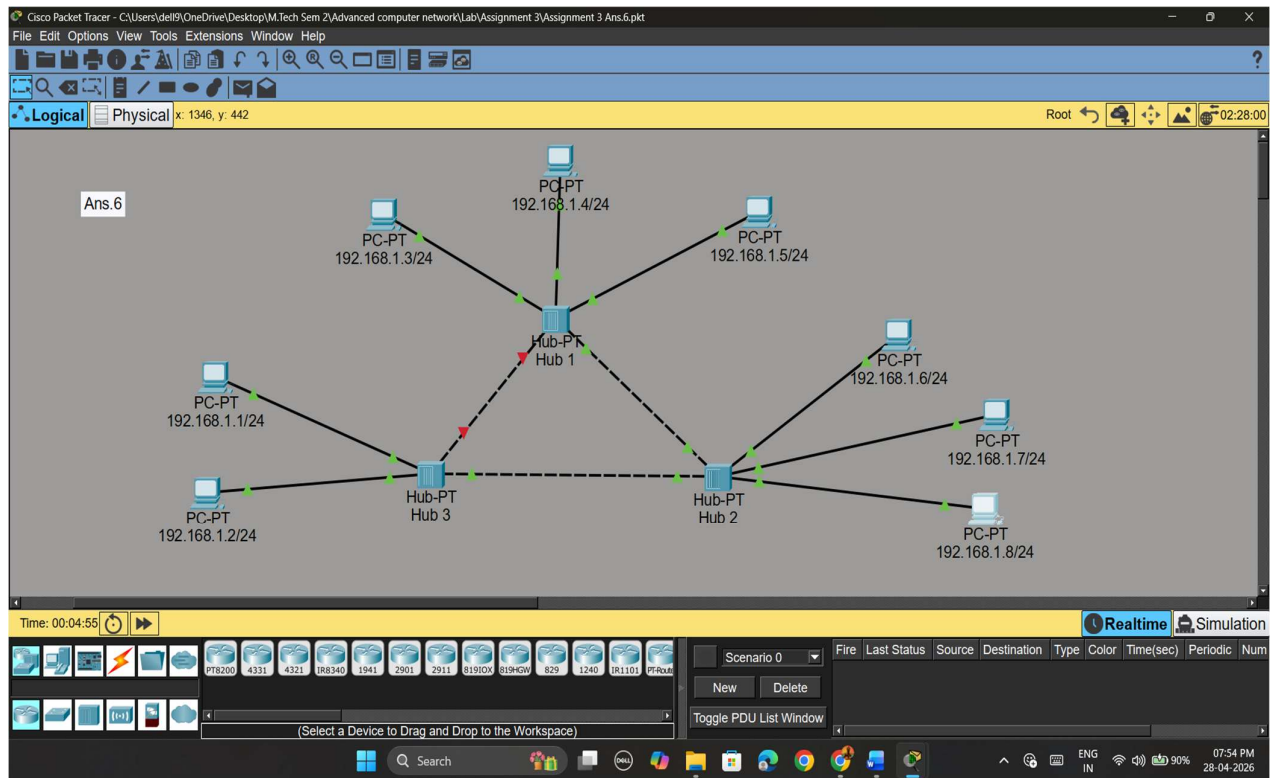
Ans.5(a)



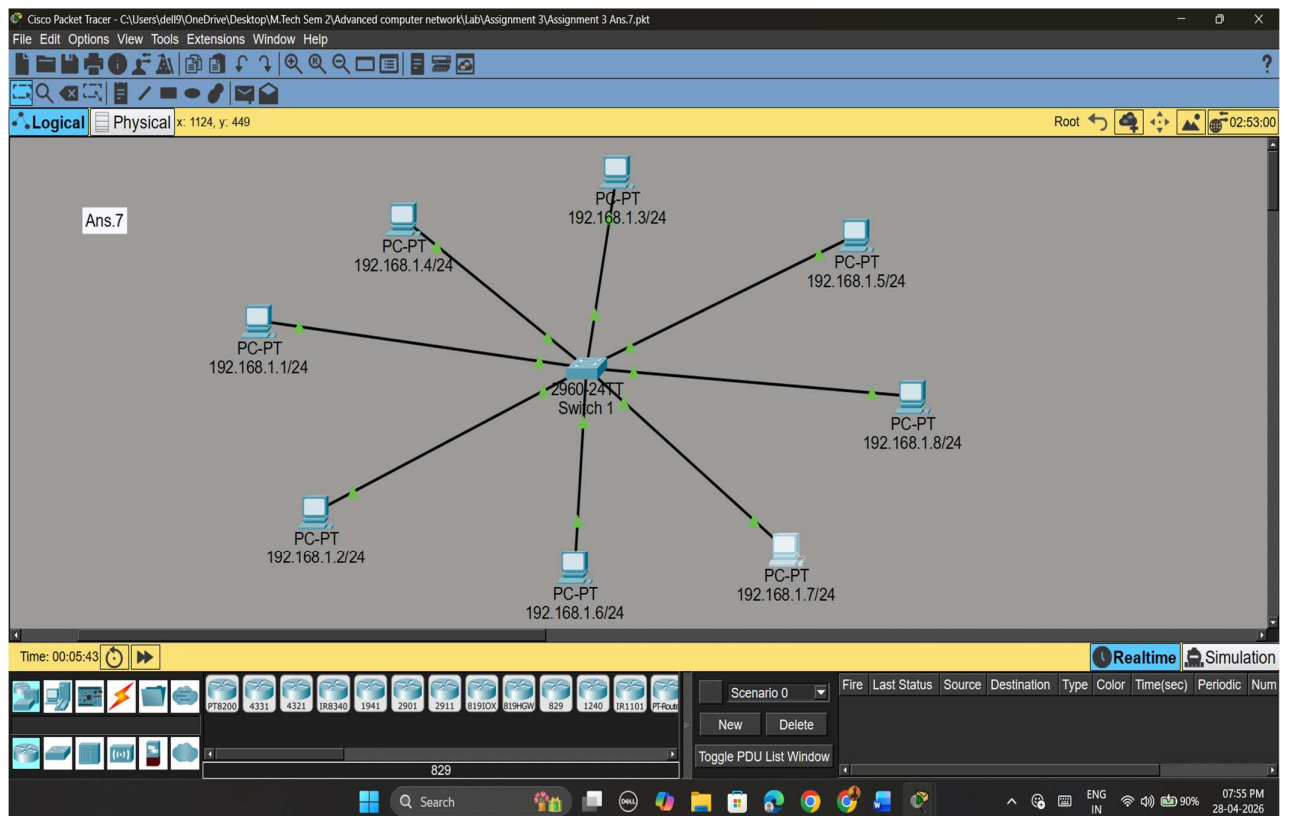
Ans.5(b)



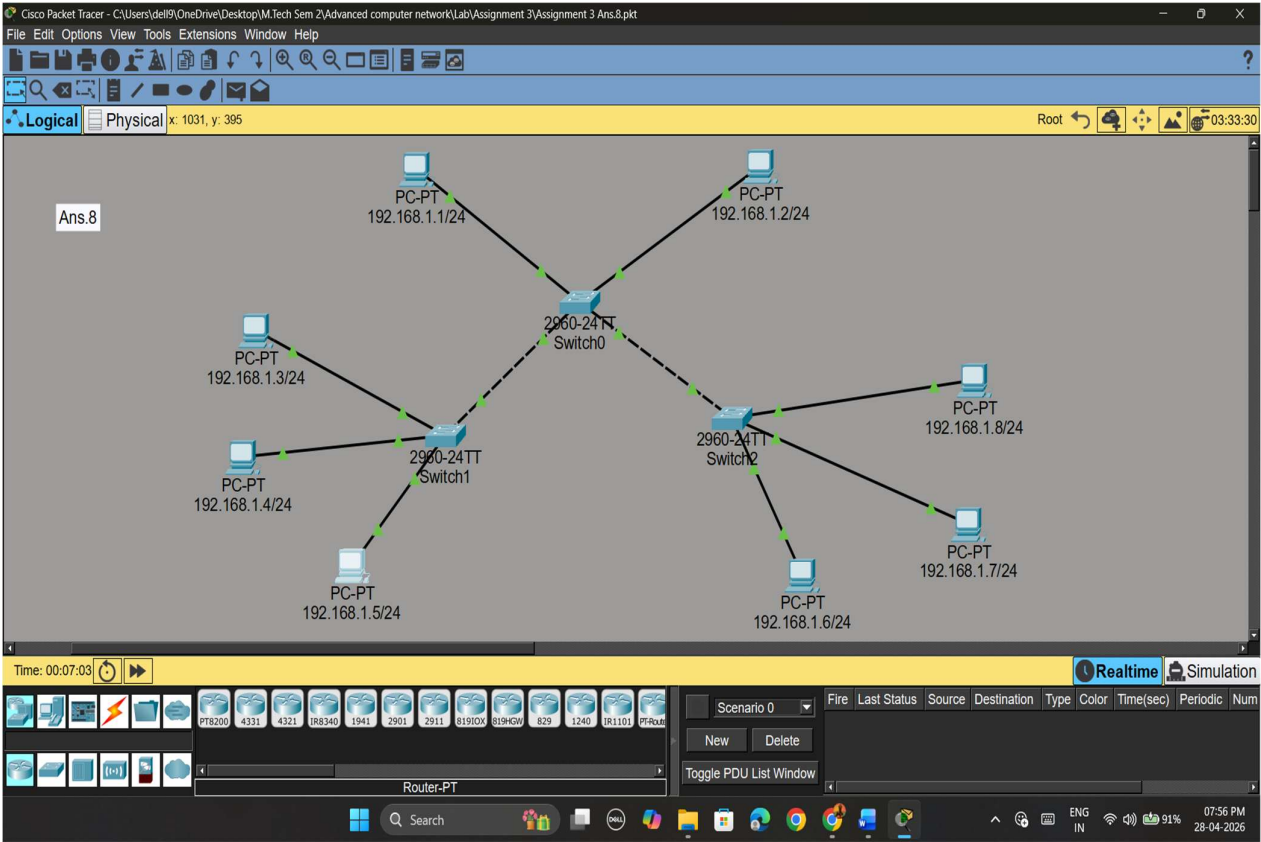
Ans.6



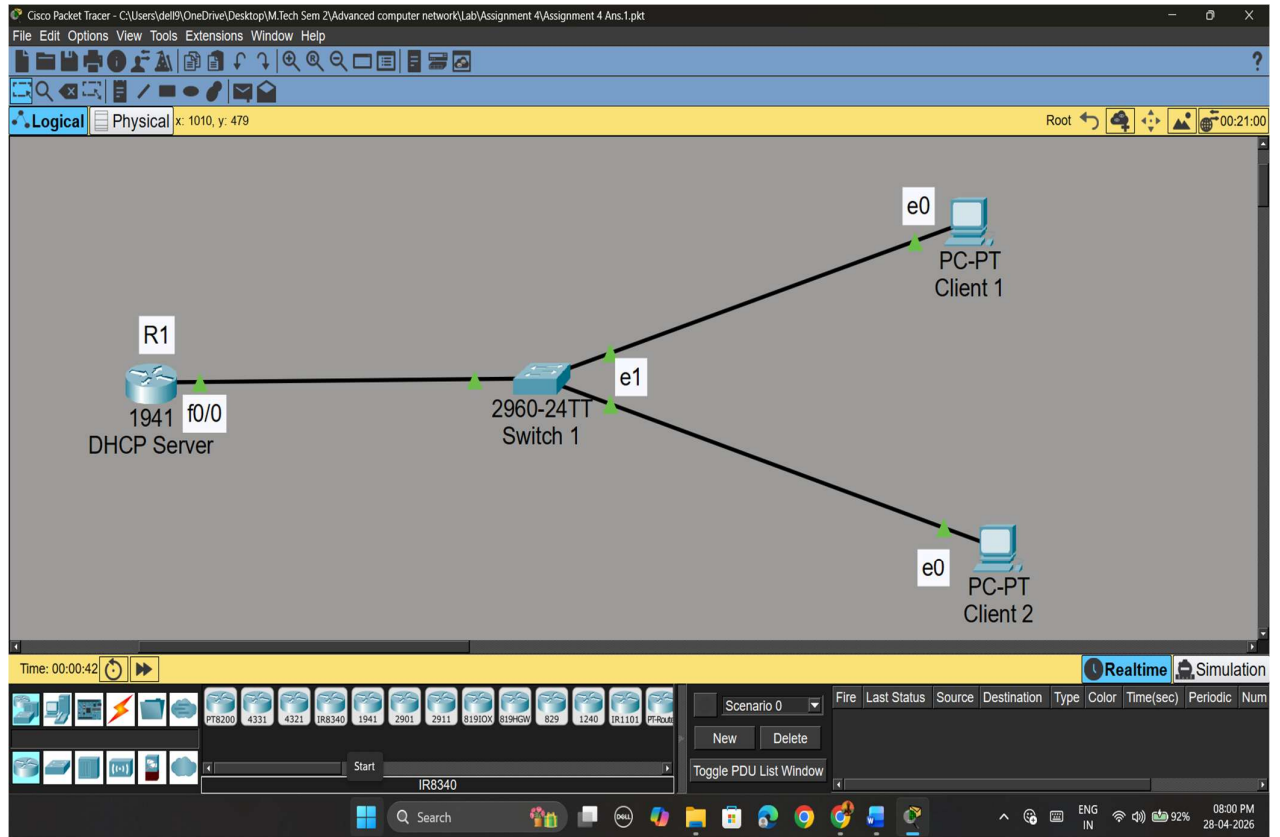
Ans.7



Ans.8

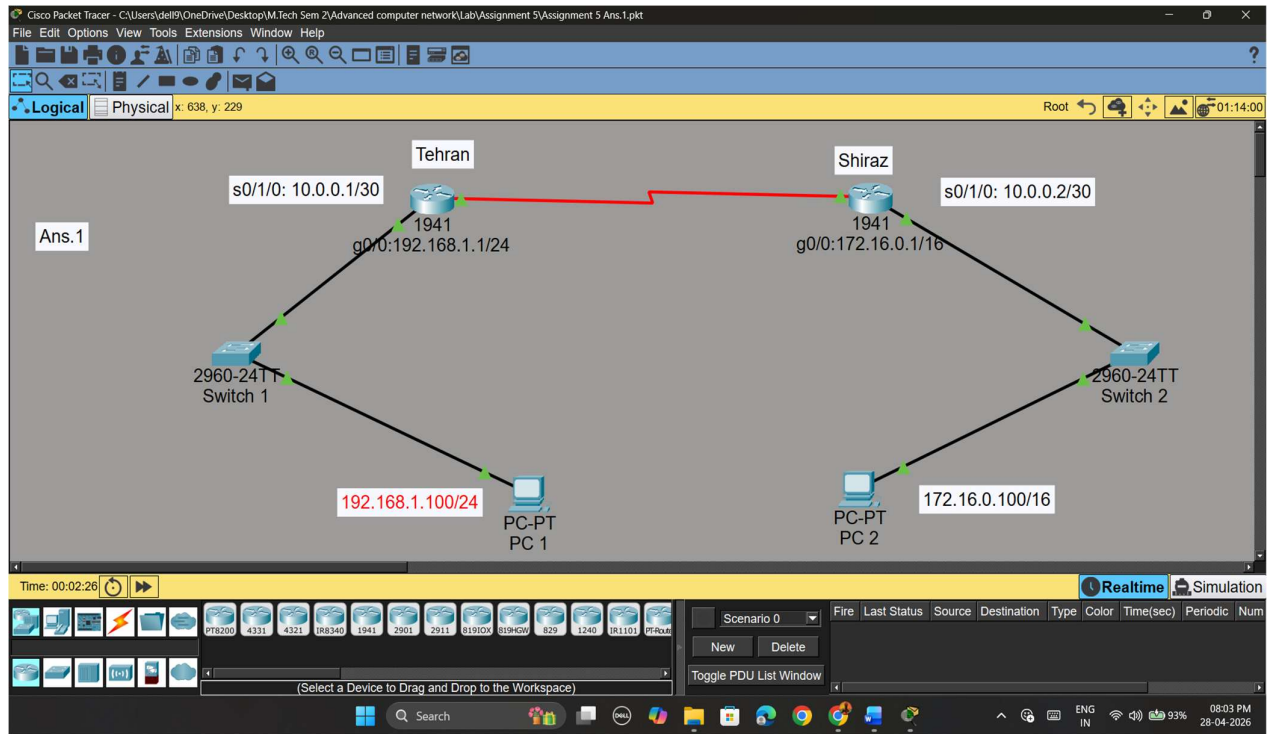


ASSIGNMENT 4

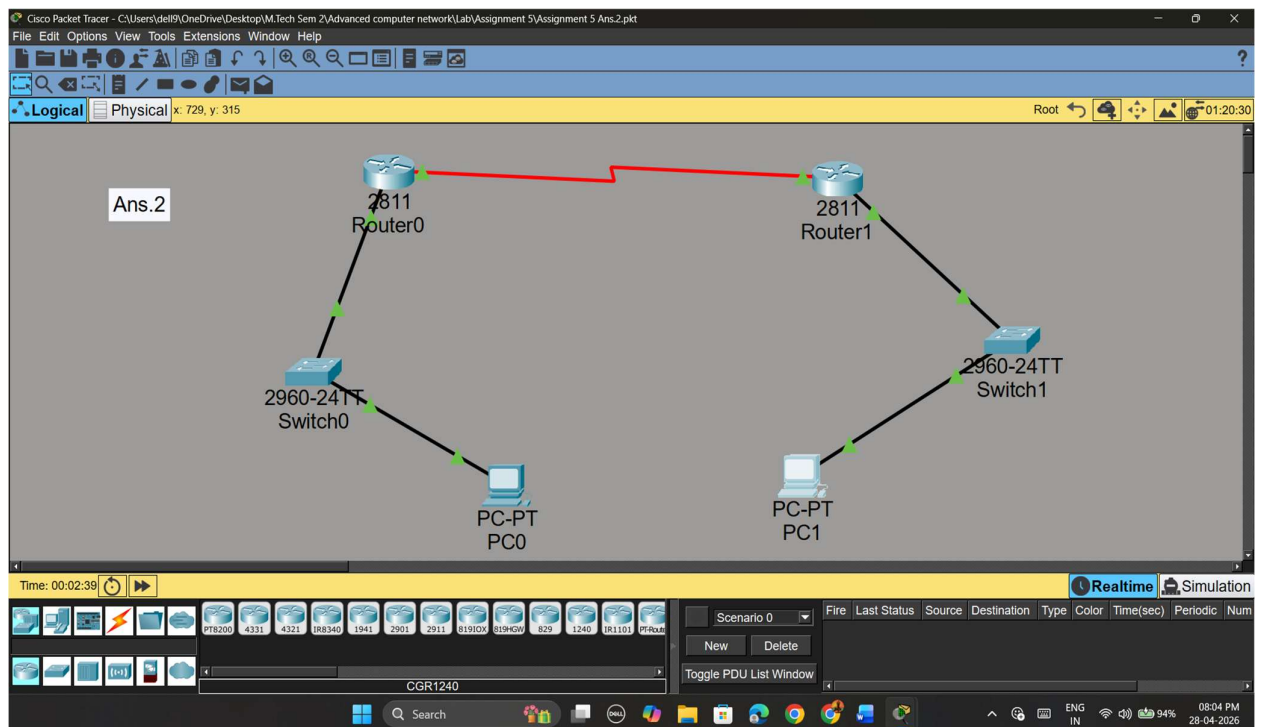


ASSIGNMENT 5

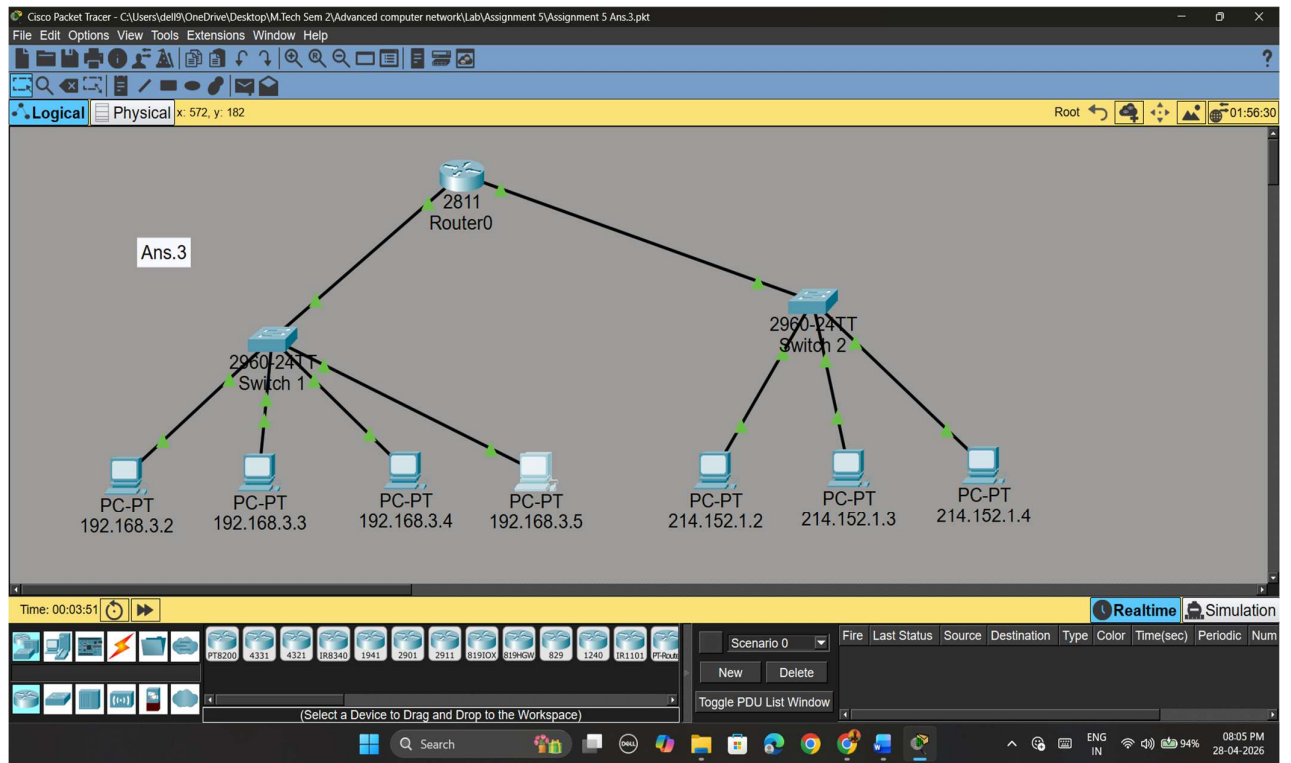
Ans.1



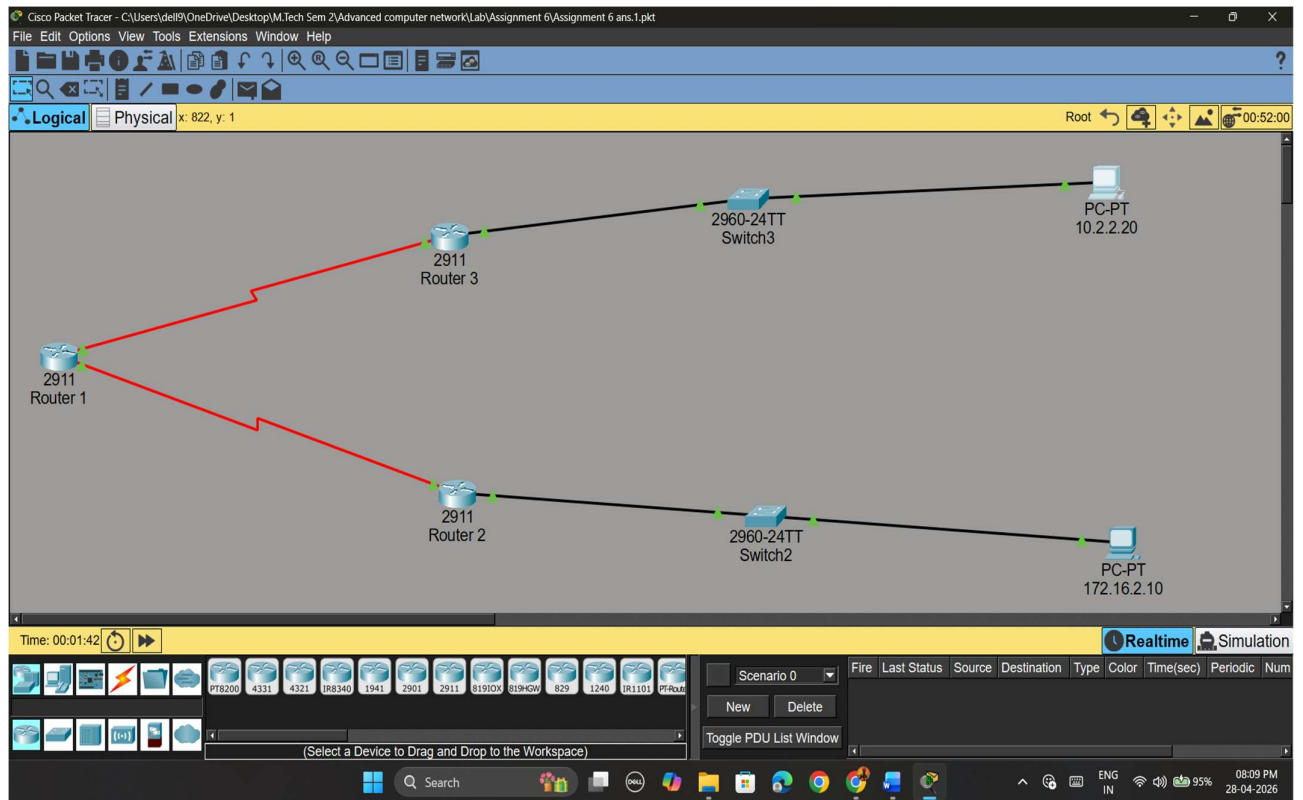
Ans.2



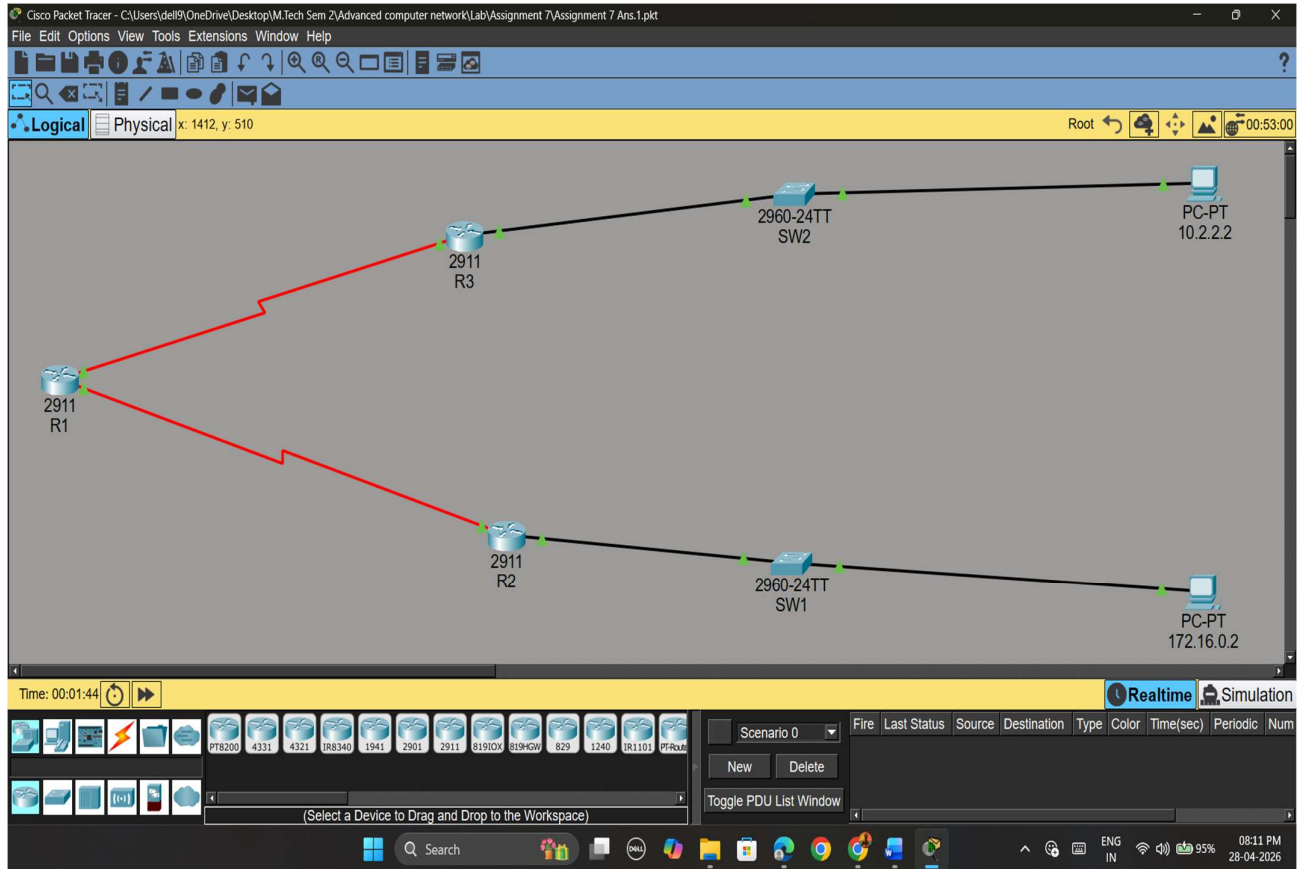
Ans.3



ASSIGNMENT 6



ASSIGNMENT 7



Additional Analytical Task (NEW)

Fill the table:

Router	Destination	Path	Cost
R1	10.2.2.0	R1→R3	64

ASSIGNMENT 8

